भारतीय प्रबंध संस्थान बेंगलूर
INDIAN INSTITUTE OF MANAGEMENT
BANGALORE

Marketing Analytics

Project Report

Topic:

"Behavioral Pattern Analysis of Loan Sanctions"

Under the guidance of

Prof. Nalini Guhesh

Group 2

Nikita Rath	2411579
Anurag Kumar	2411497
Muskan Aanand	2411520
Lavanshi	2411176

1. Context

One of our NBFC client is operating in a highly competitive and regulated environment, where effective loan recovery is critical to maintaining profitability and liquidity. As loan portfolios are expanding, traditional collection methods such as uniform reminder calls or manual follow-ups have become inefficient, costly and detrimental to customer experience.

Customers display distinct behavioural and emotional patterns during the repayment lifecycle. Some respond positively to reminders, others delay engagement, while a few react negatively to repeated outreach. From a marketing perspective, the lack of behaviour-based segmentation, message personalization and channel optimization limits the effectiveness of collection campaigns.

2. Business Objective

2.1 Overall Objective

To transition the NBFC client from a generic, high-cost manual collection model to a **data-driven, behavioral recovery framework** that optimizes **communication strategies** for each specific consumer segment while preserving customer relationships.

This approach transforms the collection process from a source of friction into a sophisticated customer lifecycle touchpoint, balancing immediate financial recovery with the long-term brand equity and customer lifetime value of the borrower.

2.2 Specific Objective

1. **Behavioural Segmentation:**
   a. ***Business Problem:*** The current single collection approach is inefficient and costly, as it fails to account for the unique behavioural and emotional triggers of different borrowers.
   b. ***Goals:*** To develop a data-driven strategy recommendation that optimizes the timing, tone and intensity of outreach for each segment, maximizing recovery rates while minimizing friction in the customer experience.

2. **Recovery Strategy Optimization**
   a. ***Business Problem:*** Uniform allocation of collection efforts results in inefficient use of resources, with high-risk accounts not receiving adequate focus and low-risk customers being over-contacted, increasing operational costs and potential brand damage.
   b. ***Goals:*** To leverage predictive risk scores and behavioural insights to prioritize accounts, optimize outreach intensity and allocate collection resources effectively in order to maximize recovery efficiency and reduce avoidable friction.

3. Analytical Objective

1. **Predictive Risk Modelling:** To develop and validate a predictive classification model that calculates a probability of default for each borrower, treating missed payments as a lead indicator of customer disengagement and credit risk.

2. **Prescriptive Behavioural Segmentation:** To perform multidimensional clustering analysis to group the loan portfolio into distinct behavioural personas, defined by the intersection of financial stability, engagement levels, and emotional state.

4. Dataset

The dataset is sourced from TVS Credit data analytics challenge (available online) capturing structured information on customer-level behavioural, transactional and interaction data across the loan repayment journey. It is designed to support customer segmentation, engagement analysis, and personalization modeling, making it suitable for marketing analytics use cases in financial services.

The raw data file (Analytics_loan_collection_dataset) used in the analysis includes:

1. **Customer demographics:** age, income, location, employment status.
2. **Loan details:** amount, tenure, interest rate, type.
3. **Repayment history:** missed payments, delays, partial payments.
4. **Interaction logs:** previous communication attempts, sentiment scores, response times.
5. **Behavioural data:** app usage, website visits, complaint history.

5. Exploratory Data Analysis (EDA)

The dataset spans a cross-sectional snapshot of **2,000 unique borrowers** across a mid-sized NBFC's active loan portfolio, covering **19 variables** across five thematic domains: demographics, loan characteristics, repayment behaviour, interaction logs and digital engagement.

The target variable likelihood of missing the next payment is near-perfectly balanced at 49.2% positive class (984 defaulters) and 50.8% negative class (1,016 non-defaulters), eliminating need for resampling or class-weight adjustments.

5.1 Initial Analysis

The average borrower is 42.5 years old, earns ₹11.1L annually, and carries a loan of ₹5.26L over a 33-month tenure at a mean interest rate of 10.6%. The average customer has missed 1.92 payment cycles and accumulated 87.4 cumulative delay days, indicating that payment stress is a pervasive not exceptional across the portfolio.

Column Name	Description	Type	Mode	Mean	Median	Min.	Max.
CustomerID	Uid for each customer	Meta					
Age	Age of the customer in yrs.	Numeric	45	42.512	43	20	65
Income	Annual income in INR	Numeric	1443064	1111400	1117180	168920	2009260
LoanAmount	Total principal amt. borrowed	Numeric	286562	526185	520010	46270	1015500
TenureMonths	Loan duration in months	Numeric	20	33.1345	33	5	61
InterestRate	Annual interest rate percentage	Numeric	10.95	10.569	10.435	6.11	15.47
MissedPayments	Number of payment cycles missed	Numeric	1	1.922	2	0	8
DelaysDays	Total cumulative delay in days	Numeric	86	87.359	86	0	179
PartialPayments	Number of partial payments made	Numeric	1	0.996	1	0	6
InteractionAttempts	Number of contact attempts made by the team	Numeric	6	4.452	4	0	9

SentimentScore	Score from past interactions	Numeric	-0.35	-0.00228	0.015	-1	1
ResponseTimeHours	Average time taken by customer to respond	Numeric	18.59	36.4907	36.75	0.55	71.99
AppUsageFrequency	Score of engagement with the mobile app	Numeric	0.68	0.49968	0.5	0	1
WebsiteVisits	Frequency of logins to the loan portal	Numeric	46	23.952	23.5	0	49
Complaints	Total formal grievances registered	Numeric	0	0.492	0	0	4

Column Name	Description	Type	Values
Location	Geographical classification	Categorical	Suburban / Urban/Rural
EmploymentStatus	Current professional status	Categorical	Salaried / Self Employed / Student / Unemployed
LoanType	Category of the loan	Categorical	Auto / Business / Education / Home / Personal
Target	Likelihood to miss the next payment	Categorical	1 (Miss), 0 (No miss)

5.2 Outlier Analysis

1. **Missed Payments:** Some customers show extreme behavior (up to 8 missed payments), which are significant predictors for the target.
2. **Delays:** DelaysDays shows a widespread (0 to 179 days), with several entries in the upper quartile indicating chronic late payers.

5.3 Correlation Analysis

A Spearman correlation matrix was computed to evaluate monotonic relationships across features. The results indicate **very weak pairwise relationships**, with the highest observed absolute correlation being only **0.069**. This weak correlation structure suggests that borrower risk is behaviourally complex rather than structurally linear.

1	+0.069	Income	:	PartialPayments
2	+0.065	Income	:	SentimentScore
3	-0.059	DelaysDays	:	PartialPayments
4	+0.057	AppUsageFrequency	:	Income
5	-0.056	Complaints	:	InterestRate

Figure: Top 5 correlated parameter combinations & their value

Note: *The dataset is clean, balanced and mostly normalized with no missing values and does not need to be heavily pre-processed.*

5.4 Derived features

Seven engineered features: *Debt-to-Income Ratio, Risk Score, Monthly Installment, Repayment Burden, Sentiment Risk Factor, Interaction Responsiveness and Digital Engagement Index* supplement raw columns and are described below.

1. **Debt-to-Income Ratio:** Measures the total leverage of the borrower. It is a critical "Financial Stress" signal.
2. **Monthly Installment:** Estimates the monthly cash outflow required to service the loan.
3. **Repayment Burden:** Captures the percentage of monthly income consumed by this loan. A high burden often precedes disengagement.
4. **Digital Engagement Index:** A normalized score (0–1) identifying whether a customer is "Tech-Savvy" or "Technologically Disconnected."
5. **Sentiment Risk Score:** An interaction feature that amplifies risk when a negative emotional state is paired with missed payments.
6. **Risk Score:** Composite metric that quantifies the "severity" of the customer's current default status.
7. **Interaction Responsiveness:** Quantifies the efficiency of the communication loop.

Features	Formula
DebtToIncomeRatio	log(LoanAmount/Income)
RiskScore	MissedPayments+(DelaysDays/30)+(Complaints*1.5)
MonthlyInstallment	log((LoanAmount * (1+ ((InterestRate*TenureMonths)/1200)))/TenureMonths)
RepaymentBurden	log(((LoanAmount * (1+ ((InterestRate*TenureMonths)/1200)))/TenureMonths)/(Income/12))
SentimentRiskFactor	log(1+MissedPayments * (1-SentimentScore))
InteractionResponsiveness	log(1+(InteractionAttempts / (ResponseTimeHours+1)))
DigitalEngagementIndex	(AppUsageFrequency + (WebsiteVisits/49))/2

Reasons for feature engineering:

1. Raw variables such as delay days and missed payments individually may not fully capture repayment volatility or stress accumulation.

2. Engagement metrics in isolation do not reflect disengagement trajectory.
3. Complaints alone do not measure friction intensity relative to account activity.
4. Static financial variables (income, loan amount) do not represent repayment strain without normalization.

6. Techniques Used, Analysis and Results

6.1 Random Forest

1. **Paramaters:** The Random Forest model was implemented to serve as the predictive engine of the recovery framework, identifying "At-Risk" signals within the customer data. **Number of trees = 18.** This specific number of trees provides enough complexity to capture non-linear interactions between behavioral features.
2. **Results:** While the model has high precision (92.6%), the recall is slightly lower (89.4%), meaning it misses about 10.6% of actual defaulters.

Evaluation results for target 1

Model	CA	F1	Prec	Recall	Spec
Random Forest	0.913	0.910	0.926	0.894	0.930

Figure: Evaluation result for our model (Random Forest)

Actual \ Predicted	0	1	Σ
0	936	70	1006
1	103	871	974
Σ	1039	941	1980

Figure: Confusion Matrix of our model (Random Forest)

3. **Implications:**
   a. A high Precision ensures that very few good borrowers are incorrectly flagged as defaulters.
   b. Because the model is highly accurate, the business can move from manual oversight to automated nudges.
   c. Regulators often look at the recall to ensure that risk is being managed effectively to protect the institution's stability.
   d. The high specificity means the system is excellent at identifying people who will pay on time.

   Overall: The integrated implication of the model is that its high predictive accuracy (91.3%) enables a transition from reactive debt collection to a proactive, automated recovery ecosystem. By balancing high precision (0.926) with strong specificity (0.930), the business can aggressively target actual defaulters to maximize ROI while simultaneously protecting the social and commercial trust of reliable borrowers.

6.2 K-Means Clustering

To determine the optimal number of 4 clusters, the experimentation involved a multi-stage iterative process using both statistical validation and business logic. This was done by **Silhouette Analysis** to measure how well each customer was assigned to their cluster relative to others, ensuring high intra-cluster similarity and clear inter-cluster separation. Finally, business feasibility was check to the results of 3, 4, and 5-cluster models.

The 4-cluster solution was selected as it provided the most actionable differentiation between high-capacity "ghosts," hostile high-risk accounts, struggling students, and cooperative safe borrowers, whereas fewer clusters over-generalized these distinct behavioral triggers.

1. Cluster C1: The High-Worth Ghost

- **Key Findings:** This cluster represents "Willful Defaulters" who possess the highest income relative to debt (DTI -1.00) but have the highest delay days (104.5). They are digitally savvy (Index 0.63) but intentionally "ghost" the company, resulting in a massive Response Time Sum of 26,011 hours.
- **Business Insights:** The cost of recovery is purely operational (reaching them) rather than financial (their ability to pay). They remain polite (0.25 sentiment) when cornered but prioritize other expenditures over loan repayment.

2. Cluster C2: The Vocal Adversary

- **Key Findings:** This is the highest risk segment (6.84) with significant hostility (-0.44 sentiment). They respond quickly (20.0 hrs) but likely use that communication to voice complaints or disputes rather than to facilitate payment.
- **Business Insights:** Their default is driven by dissatisfaction or active friction with the product or service. They are urban, salaried professionals who feel empowered to push back against collection efforts.

3. Cluster C3: The Struggling Scholar

- **Key Findings:** Comprised largely of students with the lowest digital engagement (0.38), this group is struggling to keep up but shows lower risk (4.39) than the salaried defaulters. They require the most effort (5.8 interaction attempts) to elicit a response.
- **Business Insights:** This segment lacks financial "muscle" (Moderate DTI) and experience. Their delays (70.8 days) are likely due to genuine cash flow constraints rather than a lack of intent.

4. Cluster C4: The Digital Darling

- **Key Findings:** These are the "Model Citizens" of the portfolio with the highest sentiment (0.52) and fast, reliable response times (22.7 hrs). Despite having the lowest relative income (DTI -0.64), they are consistently in the "Safe" zone (Target 0).
- **Business Insights:** This segment values their relationship with the financial institution and utilizes digital channels effectively to manage their personal loans.

7. Strategies

1. **C1: The High-Worth Ghost:** Leverage their high Digital Engagement Index (0.63) to send daily, firm WhatsApp and App-push notifications.
   Focus on the immediate impact to their high credit capacity and the legal implications of their 104-day delay.

2. **C2: The Vocal Adversary:** Assign a senior human "Dispute Specialist" rather than a standard collector to address their -0.44 Sentiment Score.
   Empathetic and solution-oriented; prioritize resolving their complaints or service friction before requesting payment.

3. **C3: The Struggling Scholar:** Offer "Micro-Payment" plans or temporary interest waivers tailored for their student status and lower digital engagement.
   Supportive and educational; focus on long-term credit health and financial counseling rather than immediate demand.

4. **C4: The Digital Ace:** Immediately remove from all recovery workflows to avoid brand damage from over-contact.
   Transition to marketing; provide appreciative messages, pre-approved top-up loans, or loyalty rewards based on their 0.52 Positive Sentiment.

8. Conclusion

This study shows that loan default behavior is driven by diverse financial conditions, engagement levels, and emotional responses, making uniform collection strategies ineffective. By combining a high-accuracy predictive model (Random Forest) with behavioral segmentation (K-Means), the NBFC can identify at-risk borrowers early and tailor recovery actions to distinct customer personas. This data-driven approach improves recovery efficiency, reduces operational costs, minimizes unnecessary customer friction, and preserves long-term relationships and brand value by shifting from reactive collections to proactive, personalized interventions.

9. Future Directions

1. **Real-Time Risk Scoring:** Continuously update borrower risk using live behavioral and payment data for timely interventions.

2. **Advanced Modeling Techniques:** Apply models like XGBoost or deep learning to capture complex patterns and improve prediction accuracy.

3. **Omnichannel Communication Optimization:** Use analytics to select the best channel, timing, and frequency of outreach for each customer segment.

4. **Early Distress Detection:** Incorpor external financial and behavioral indicators to identify repayment stress before missed payments occur.

APPENDIX

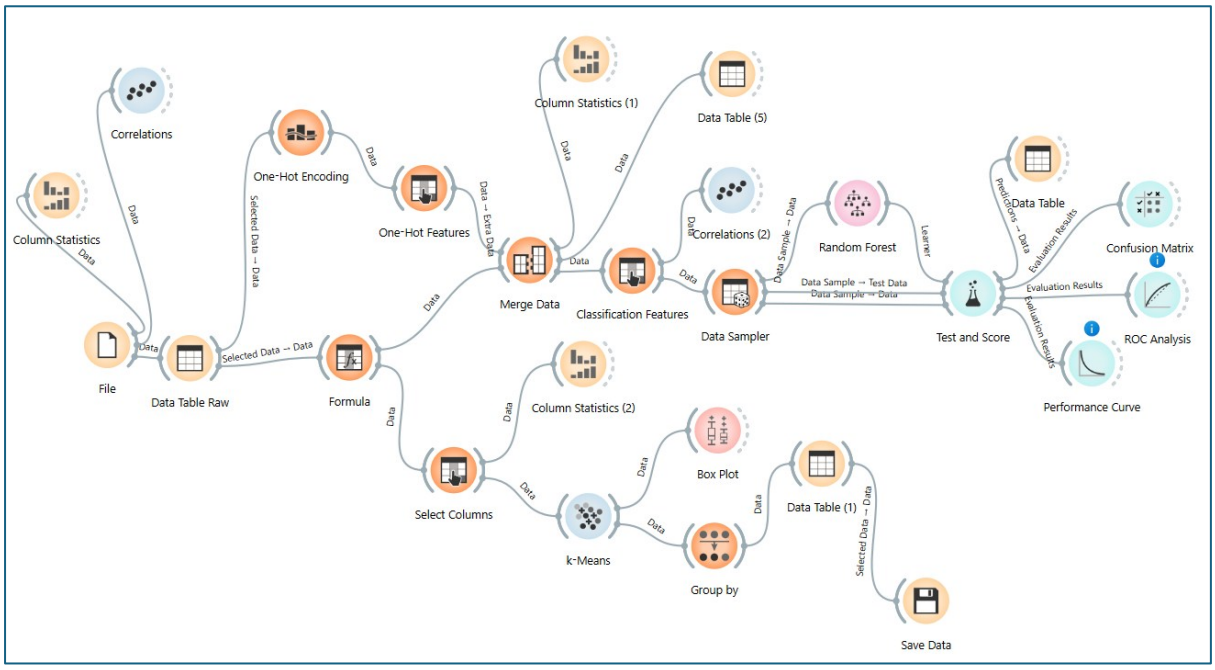

Figure: Orange Workflow